

Statistics: Normal Distribution

Hsueh-Tse, Cheng
b11303063@ntu.edu.tw

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1 Normal distribution (常態分布)

1.1 Properties of normal curves

1. The standard normal distribution, based on z-scores = $\frac{score - mean}{SD}$. Its pdf (probability density function) is:

$$f(z) = \frac{1}{\sqrt{2\pi}} e^{-\frac{z^2}{2}}$$

2. Other normal distribution's pdf:

$$f(x) = \frac{1}{\sigma\sqrt{2\pi}} e^{-\frac{(x-\mu)^2}{2\sigma^2}}$$

3. Any normal distribution's variance is larger than 0, it cannot be zero.
4. A normal curve is **continuous** and asymptotic to the x-axis.
5. μ and σ^2 are two parameters of the normal distribution.

1.2 Central Limit Theorem (中央極限定理)

As the sample size (N) increases, the **sampling distribution** of sample means ($\bar{X}$):

1. decreases in variability
2. the shape of the distribution approaches a normal curve
3. the expected mean of sample means = population mean

$$E(\bar{X}) = \mu$$

→ This is because $\bar{X}$ is an **unbiased** estimator of population μ

1.3 Standard error

Standard error is the standard deviation of the sampling distribution of a sample mean ($\bar{X}$). According to the Central Limit Theorem, standard error is denoted as:

$$\sigma_{\bar{X}} = \frac{\sigma}{\sqrt{n}}$$